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ADMINISTRATION OF ASPARTAME IN NON-INSULIN-DEPENDENT DIABETICS

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A study was designed to determine the effect of the consumption of the nutritive sweetener aspartame on non-insulin-dependent diabetics. Forty-three adult diabetics between the ages of 21 and 70 completed a 90-day study; all were diabetics whose conditions were managed by diet and/or hypoglycemic agents. Participants in the blind study were instructed to continue their usual diet and to take two capsules of an assigned preparation three times daily with meals, either the aspartame or the placebo. The 1.8 g of aspartame administered is approximately three times the expected daily consumption of aspartame if used as a sweetener to replace sugar. Throughout the study subjects were examined for (1) symptoms of intolerance, (2) fasting plasma phenylalanine levels exceeding 4 mg/100 ml, and (3) deterioration of diabetic control. At the conclusion of the study subjects exhibited no symptoms that could be traced to the administration of aspartame or the placebo, and diabetic control was unaffected by the chronic administration of these substances. Aspartame seems to be well tolerated by non-insulin-dependent diabetics.

INTRODUCTION

The nutritive sweetening agent aspartame (L-aspartame-L-phenylalanine methyl ester) is approximately 180 times as sweet as sugar with 0.5% the caloric value for an equivalent amount of sweetening. It is anticipated that aspartame might therefore become widely used as a sugar substitute. In those persons with metabolic disorders that require limited use of sugar, such as diabetes mellitus, aspartame would be of special value.

This report is a summary of our study that deals with the tolerance of non-insulin-dependent diabetics to aspartame.

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OBJECTIVES

The present study was designed to determine whether non-insulin-dependent diabetic subjects could consume 1.8 g aspartame daily for 90 days (a) without signs or symptoms of intolerance occurring, (b) without fasting plasma phenylalanine (PA) levels exceeding 4 mg/100 ml, and/or (c) without deterioration in diabetic control. The amount of aspartame administered constituted an intake of approximately three times the expected adult daily consumption of aspartame when used as a sweetener.

MATERIAL

The subjects involved in this study were adults between the ages of 21 and 70 whose disease was being managed by diet and/or oral hypoglycemic agents. They did not receive insulin. The subjects were presumed to have a normal ability to metabolize phenylalanine if their noon 4-hr fasting PA level did not exceed 4 mg/100 ml.

Prior to acceptance into the study, each subject was given a complete physical examination. Only those individuals whose results for the following tests were within normal limits were accepted into the study: complete blood count, pregnancy test (women), partial thromboplastin time (PTT), BUN, creatinine, bilirubin, plasma phenylalanine, plasma tyrosine. In addition, the following tests were also done at screening, although elevated levels did not preclude the patient's entry into the study: SGOT, SGPT, LDH, alkaline phosphatase, uric acid, cholesterol (total and esters), triglycerides, fasting blood sugar, oral glucose tolerance test (selected cases), and urinalysis.

Complete eye examinations, including tonometry, fundoscopy, and slit lamp examination, were done prior to and on completion of the study.

The study design was double blind with the subjects randomly assigned to receive aspartame or matching placebo capsules. They were instructed to continue their usual diet and to take two capsules of the assigned study preparation three times daily with meals.

PROCEDURE

Initial Screening Period

A complete history and physical examination was performed 1 wk prior to the administration of aspartame (or placebo). The laboratory procedures were performed as outlined above, as was a complete ophthalmologic examination. The patient's diet and any concurrent medication was also duly noted and recorded.

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Test Period

During each week of the 13-wk test period, the randomly assigned volunteers in this double-blinded study were given two capsules three times daily with meals. Those receiving aspartame received 0.6 g three times daily for a total of 1.8 g daily during this period of testing. Laboratory parameters were evaluated at the end of week 1, 3, 5, 7, 9, and 11.

Follow-up Period

Final evaluations, both clinical and chemical, were made on all subjects who had completed the entire testing period (week 14). Another complete ophthalmologic examination was also performed and compared with the examination performed during the screening period.

RESULTS

Sixty-nine subjects completed the study. Table 1 indicates the distribution. Six other participants were lost to follow-up or were discontinued for medical reasons not related to the study.

Effects of Aspartame on Diabetic Control

In the group of 43 patients reported by one of the investigators (Stern) the following effects were noted:

1. Excellent control—This is defined as those patients in whom the majority of their fasting blood sugars were less than 110 mg%: aspartame, 7 patients; placebo, 11 patients.
2. Good control—This includes those subjects whose fasting blood sugars were greater than 110 mg% but less than 130 mg% for the majority of their tests: aspartame, 7 patients; placebo, 3 patients.
3. Poor control—This is defined as those patients in whom the majority of their blood sugars were greater than 130 mg%: aspartame, 9 patients; placebo, 6 patients.

TABLE 1. Study Population: Non-insulin-dependent Diabetics

	Aspartame		Placebo		
Study of	Male	Female	Male	Female	Total
Bleicher	5	9	0	12	26
Stern	4	18	3	18	43
Total	9	27	3	30	69

There was no evidence in this group that aspartame exerted an influence on diabetic control. the patients in this study continued to show the same degree of diabetic control during the period capsules were taken and during the follow-up period as they had exhibited during the screening period.

The remaining 26 patients followed by the other investigator (Bleicher) had their plasma glucose data statistically analyzed. The fasting glucose determinations were reviewed for the study and placebo groups at weeks 1, 5, 9, and 14. The mean maxima for glucose and other tests are presented with standard error on Table 2 for treatment and placebo groups as well as the *t* test results.

As was reported in the other 43 patients, there was no evidence that aspartame exerted an influence on diabetic control. Patients in this study continued to show the same degree of diabetic control during the period capsules were taken and during the follow-up period as they had exhibited during the screening period.

Patients were carefully questioned concerning compliance in taking aspartame or placebo. In addition, counts were made of the number of capsules present in each bottle when they were returned and the amounts of capsules that were gone corresponded to the number that had theoretically been consumed. Although one cannot state with 100% certainty that all capsules were consumed, it would seem to be a reasonable assumption in view of these figures.

In respect to other drugs, patients took their usual antihypertensives, tranquillizers, etc., but these medications in general were constant for each patient throughout the entire study. Furthermore, there was no difference in terms of random patient assignments to each group so that the distribution of medications to each category would appear to be equivalent.

Effects of Aspartame on Weights and Other Measurements

There appeared to be no significant differences between the two treatment groups in mean weight, systolic blood pressure, diastolic blood pressure, or pulse either initially or at the end of the study (Table 3).

Effects of Aspartame on Laboratory Values

The means of various laboratory determinations in the aspartame and in the placebo group means of 2-tied *t* tests revealed that LDH was significantly higher in the placebo group in the initial tests. Hemoglobin was somewhat higher in the placebo group but it was a difference of borderline significance.

Among the final values, the creatinine level was significantly high in the placebo group. Hemoglobin, SGPT, and cholesterol differences were of

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TABLE 2. Non-insulin-dependent Diabetics: Mean 8:00 a.m. Phenylalanine, Tyrosine, and Glucose Levels (Bleicher Group)

	Week			
	1	5	9	14
		Serum phenylalanine levels		
Aspartame	1.47 ± 0.166 (9)[a]	1.71 ± 0.166 (9)	1.76 ± 0.158 (8)	1.64 ± 0.165 (9)
Placebo	1.20 ± 0.210 (8)	1.87 ± 0.130 (7)	1.73 ± 0.149 (8)	1.74 ± 0.161 (11)
		Tyrosine		
Aspartame	1.00 ± 0.174 (8)	0.80 ± 0.122 (9)	1.30 ± 0.134 (8)	1.12 ± 0.212 (9)
Placebo	1.04 ± 0.074 (8)	1.06 ± 0.152 (7)	1.31 ± 0.078 (8)	1.35 ± 0.079 (11)
		Fasting glucose		
Aspartame	115.7 ± 13.94 (10)	116.2 ± 10.76 (10)	136.5 ± 16.94 (9)	134.9 ± 20.73 (10)
Placebo	88.9 ± 8.61 (11)	107.2 ± 11.10 (11)	117.7 ± 13.56 (10)	95.5 ± 11.17 (11)
t	1.64	0.58	0.87	1.67
p	> 0.10	> 0.50	> 0.40	> 0.10

[a] Number in parentheses is the number of subjects.

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TABLE 3. Non-insulin-dependent Adult Diabetics Weight and Other Measurements

Measurement	Aspartame		Placebo	
	Initial	Final	Initial	Final
Systolic BP	144.5 ± 3.43 (34)	145.2 ± 3.78 (31)	140.7 ± 4.62 (31)	143.1 ± 4.87 (31)
Diastolic BP	82.6 ± 1.84 (34)	84.7 ± 2.09 (31)	81.4 ± 2.10 (31)	83.6 ± 2.30 (31)
Pulse	79.3 ± 0.90 (34)	79.8 ± 1.18 (31)	80.3 ± 1.33 (31)	81.4 ± 1.69 (31)
Weight	176.8 ± 6.67 (35)	177.0 ± 6.42 (33)	175.3 ± 5.92 (29)	175.1 ± 5.84 (29)

TABLE 4. Non-insulin-dependent Diabetics Laboratory Values

Measurement	Aspartame		Placebo	
	Initial	Final	Initial	Final
Hgb	14.4 ± 0.022 (35)	13.9 ± 0.18 (34)	13.8 ± 0.30 (28)	13.4 ± 0.22 (27)
Hct	43.2 ± 0.49 (35)	41.6 ± 0.51 (33)	41.8 ± 1.10 (28)	40.5 ± 0.61 (27)
WBC (X 1,000)	6.83 ± 0.363 (35)	6.040 ± 0.332 (33)	7.010 ± 0.475 (28)	5.684 ± 0.261 (27)
PTT	52.9 ± 2.82 (34)	63.0 ± 4.04 (33)	57.5 ± 3.20 (27)	64.1 ± 4.24 (27)
Prothrombin time	11.9 ± 0.11 (34)	12.1 ± 0.10 (33)	12.2 ± 0.11 (26)	12.2 ± 0.14 (27)
Bilirubin[a]				
Direct	0.068 ± 0.009 (15)	0.078 ± 0.012 (22)	0.053 ± 0.009 (19)	0.067 ± 0.011 (19)
Indirect	0.33 ± 0.033 (23)	0.34 ± 0.032 (22)	0.34 ± 0.050 (19)	0.31 ± 0.034 (19)
T4	7.08 ± 0.481 (35)	6.74 ± 0.413 (33)	6.67 ± 0.432 (28)	6.37 ± 0.392 (27)
Uric acid	4.90 ± 0.280 (35)	4.43 ± 0.286 (33)	4.75 ± 0.332 (28)	4.10 ± 0.265 (27)
Creatinine	1.08 ± 0.456 (35)	0.84 ± 0.041 (33)	1.10 ± 0.048 (28)	0.96 ± 0.040 (27)
Cholesterol	191 ± 8.3 (35)	186 ± 9.7 (33)	201 ± 11.6 (28)	215 ± 12.1 (27)
Triglycerides	152 ± 25.0 (35)	134 ± 16.7 (33)	137 ± 12.5 (28)	122 ± 10.1 (27)
BUN	144.6 ± 0.49 (34)	13.4 ± 0.48 (32)	15.6 ± 0.67 (29)	13.9 ± 0.56 (30)
SGOT	50.5 ± 5.96 (34)	49.2 ± 4.46 (32)	39.4 ± 4.16 (29)	40.3 ± 2.74 (29)
Alkaline phosphatase[b]	23.2 ± 1.89 (23)	27.9 ± 2.39 (22)	21.9 ± 1.60 (19)	28.6 ± 6.25 (19)
LDH[b]	82.3 ± 3.17 (23)	80.3 ± 4.59 (22)	91.0 ± 2.73 (19)	82.9 ± 2.26 (19)
SGPT[b]	11.3 ± 1.90 (23)	10.2 ± 1.40 (22)	7.3 ± 0.53 (19)	8.9 ± 1.25 (19)

[a]Stern data only. Bleicher study did not include both direct and indirect bilirubin.
[b]Stern data only. Bleicher values in noncomparable units.

doubtful borderline significance and the initial LDH difference did not persist.

The laboratory values of individual patients fluctuated but no new abnormalities developed during the study among the subjects on aspartame or those on placebo in any of the tests (Tables 4 and 5).

Effects of Aspartame on Plasma Phenylalanine (PA) and Tyrosine (T) Levels

Plasma PA, T, and the PA/T ratio are plotted on Figs. 1–3 comparing the aspartame and placebo groups. It is obvious that no group showed appreciable changes over the course of the study, nor did the groups differ from one another to any extent.

Ophthalmologic Effects

Over the 14 wk of the study, no changes from base-line observations were noted.

Electrocardiographic Effects

No changes from base-line observations were noted over the 14 wk of the study.

One patient (L.A.D., no. 62) in the aspartame group studied by Stern was dropped from the study in the 11th week when a subtotal gastrectomy was necessitated because of the presence of a reticulum cell sarcoma. A consultant pathologist reviewed the slide of the stomach and nodes of this patient as well as slides of stomach section and regional nodes from rats that had consumed large doses of aspartame for 2 yr. He found no evidence of adverse effects on the gastric mucosa or other gastric tissues and no evidence of reticuloendothelial cell hyperplasia in the lymphoid aggregates adjacent to the stomach in the human or rat specimens, reinforcing his opinion that the occurrence was unrelated to aspartame ingestion. Furthermore, toxicologic studies with the rat, hamster, and dog in which massive amounts of aspartame were administered over a 2-yr period revealed no evidence of treatment-related neoplastic change. It is therefore the opinion of both the investigator and the pathologist that the detection of this neoplasm was fortuitous and coincidental. Another patient (in the Bleicher group) was found to be pregnant in the 11th week of the study. She was dropped from the study and was, unfortunately, lost to follow-up.

Mild gastrointestinal complaints were the most common discomforts observed in this group of patients (Table 6).

SUMMARY

This study involved the administration of aspartame and placebo to randomly assigned non-insulin-dependent diabetic subjects who submitted

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TABLE 5. Non-insulin-dependent Diabetics: Mean Maxima[a] on Treatment by Treatment Group in all Parameters[b]

	Aspartame			Placebo			*t* test		
Test	*n*	Mean	SEM	*n*	Mean	SEM	*n*	t_s	Result
Hgb	15	12.98	0.888	13	13.13	0.537	26	0.139	$p > 0.50$
Hct	15	42.8	0.712	13	42.4	0.971	26	0.338	$p > 0.50$
WBC	15	7.56	1.092	13	7.24	0.676	26	0.240	$p > 0.50$
Polys.	15	63.5	2.46	13	63.1	2.29	26	0.118	$p > 0.90$
Lymphocytes	15	45.4	1.90	13	46.1	2.42	26	0.230	$p > 0.50$
PTT	15	50.86	2.60	13	47.36	3.510	26	0.651	$p > 0.50$
Protein	15	14.58	0.763	13	14.41	0.688	26	0.163	$p > 0.50$
Bilirubin, total	15	0.48	0.022	13	0.68	0.066	26	3.055	$p < 0.01$
T4	15	5.29	0.411	13	5.86	0.434	26	0.952	$p > 0.20$
LDH	15	247.3	14.60	13	224.7	11.60	26	1.186	$p > 0.20$
SGPT	15	24.5	4.63	13	24.8	2.53	26	0.054	$p > 0.90$
Uric acid	15	6.60	0.505	13	6.75	0.375	26	0.232	$p > 0.50$
Creatinine	15	1.12	0.063	13	1.14	0.096	26	0.179	$p > 0.50$
Cholesterol, total	15	236.5	9.88	13	259.2	14.53	26	1.322	$p > 0.10$
Triglyceride	15	228.4	86.92	13	170.0	27.39	26	1.497	$p > 0.10$
Phenylalanine	15	2.67	0.163	13	2.70	0.121	26	0.144	$p > 0.50$
Tyrosine	15	1.54	0.120	13	1.56	0.087	26	0.127	$p > 0.50$
Weight	15	197.4	10.70	13	189.5	7.92	26	0.578	$p > 0.50$
Glucose	15	147.5	15.05	13	132.4	9.82	26	0.812	$p > 0.50$
SGOT	15	40.3	5.42	13	39.2	2.65	26	0.174	$p > 0.50$
BUN	15	16.4	0.95	13	22.0	3.16	26	1.804	$p > 0.50$
Alkaline phosphatase	15	84.2	7.62	13	90.3	8.13	26	0.547	$p > 0.50$
Systolic BP	15	149.3	4.42	13	156.1	6.36	26	0.897	$p > 0.20$
Diastolic BP	15	95.1	3.66	13	98.1	4.58	26	0.517	$p > 0.50$
Pulse	15	88.3	1.43	13	91.6	2.70	26	1.121	$p > 0.20$

[a]Maxima occurring on lead-out day of trial were included with above.
[b]Based on Bleicher's data.

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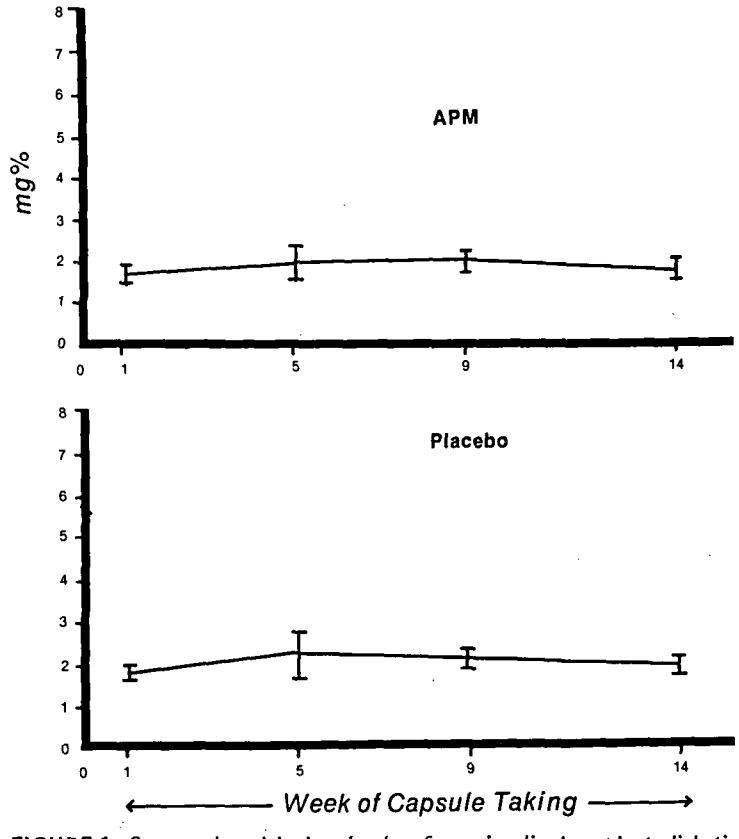

FIGURE 1. Serum phenylalanine levels of non-insulin-dependent diabetics administered aspartame (APM) and placebo. Samples were drawn at noon; sample mean with 95% confidence limits.

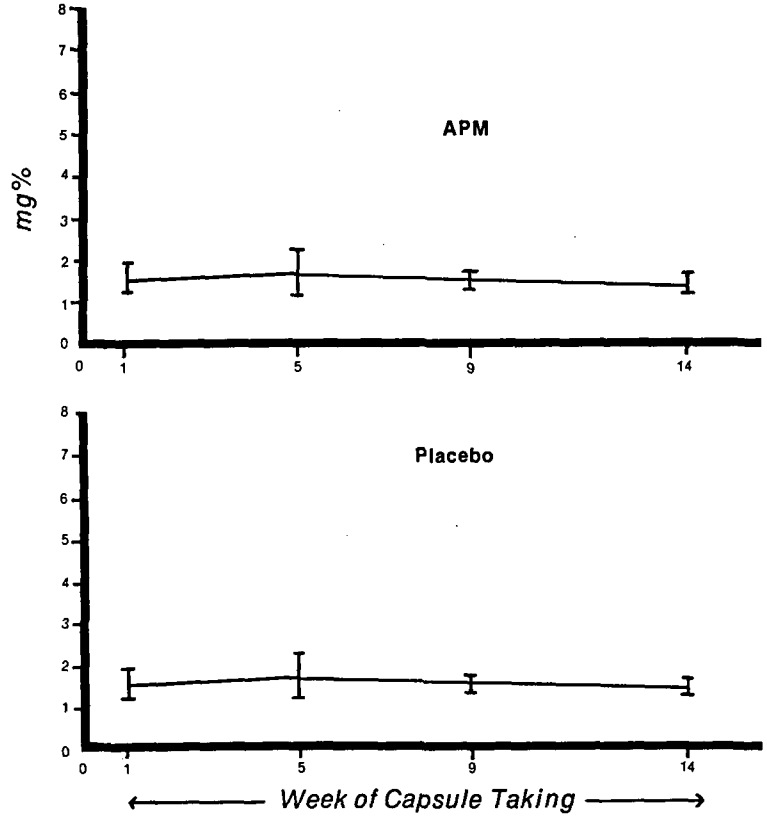

FIGURE 2. Serum tyrosine levels of non-insulin-dependent diabetics administered aspartame (APM) and placebo. Samples were drawn at noon; sample mean with 95% confidence limits.

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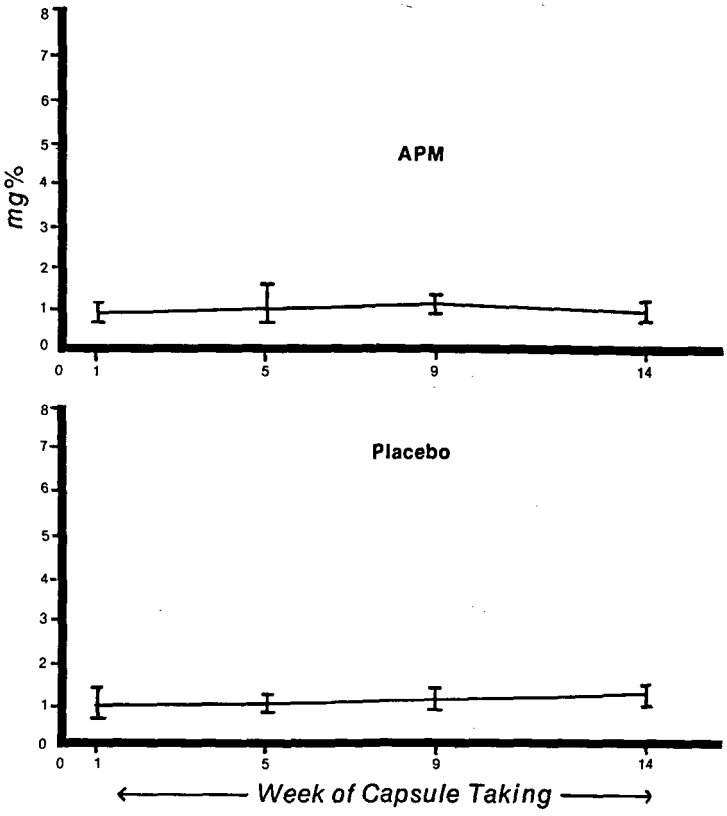

FIGURE 3. Serum phenylalanine/tyrosine ratio of non-insulin-dependent diabetics administered aspartame (APM) and placebo. Samples were drawn at noon; sample mean with 95% confidence limits.

TABLE 6. Complaints of Non-insulin-dependent Diabetics

	Incidence of symptoms	
Symptoms	Aspartame group ($n = 36$)	Placebo group ($n = 33$)
Cramps	0	2
Nausea	1	1
Constipation	2	5
Diarrhea	3	0
Loss of appetite	1	1
Nervousness	2	2

voluntarily to this double-blinded study. Subjects exhibited no symptoms that could be attributed to the administration of aspartame or placebo, and diabetic control was unaffected by the chronic administration of these substances. The various laboratory parameters that were evaluated prior to and at the end of the study showed no variations that could be attributable to aspartame, and the occurrence of sarcoma in case no. 62 was undoubtedly a fortuitous one.

In summary, aspartame appears to be well tolerated by those non-insulin-dependent diabetic subjects who participated in this 14-wk study. Concerning the impact of this negative study on long-term effects, one cannot project with confidence beyond the duration of this study.

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